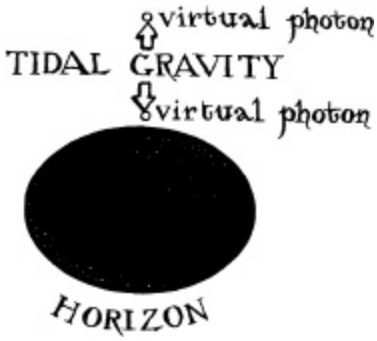
a time), living momentarily on fluctuational energy borrowed from neighboring regions of space, and that then annihilate and disappear, giving their energy back to the neighboring regions. For electromagnetic vacuum fluctuations, the virtual particles are *virtual photons;* for gravitational vacuum fluctuations, they are *virtual gravitons.*[12]



12.2 The mechanism of black-hole evaporation, as viewed by someone who is falling into the hole. *Left:* A black hole's tidal gravity pulls a pair of virtual photons apart, thereby feeding energy into them. *Right:* The virtual photons have acquired enough energy from tidal gravity to materialize, permanently, into real photons, one of which escapes from the hole while the other falls toward the hole's center.

The manner in which vacuum fluctuations cause black holes to evaporate is depicted in Figure 12.2. On the left is shown a pair of virtual photons near a black hole's horizon, as viewed in the reference frame of someone who is falling into the hole. The virtual photons can separate from each other easily, so long as they both remain in a region where the electromagnetic field has momentarily acquired positive energy. That region can have any size from tiny to huge, since vacuum fluctuations occur on all length scales; however, the region's size will always be about the same as the wavelength of its fluctuating electromagnetic wave, so the virtual photons can move apart by only about one wavelength. If the wavelength happens to be about the same as the hole's circumference, then the virtual photons can easily separate from each other by a quarter of the circumference, as shown in the figure.

Tidal gravity near the horizon is very strong; it pulls the virtual photons apart with a huge force, thereby feeding great energy into them, as seen by the infalling observer who is halfway between the photons. The increase in photon energy is sufficient, by the time the photons are a quarter of a horizon circumference apart, to convert them into real long-lived photons (right half of Figure 12.2), and have enough energy left over to give back to